

Kuldeep Tanwar

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Portfolio | LinkedIn | GitHub | LeetCode | GFG

PROFESSIONAL SUMMARY

Full-stack Engineer with 2.7 years building **scalable, production-grade systems** in Next.js, NestJS, FastAPI, and cloud-native AWS environments. Skilled in **backend optimization**, Redis caching, async architecture, and system design to improve performance, reliability, and infrastructure efficiency. Experienced in **AI-integrated development** using Claude API, OpenAI APIs, GPTZero, and LLM pipelines to automate workflows at scale. Adept at **AWS-native architecture**, performance tuning, and translating complex engineering into measurable outcomes.

EXPERIENCE

Software Development Engineer (Next.js, NestJS, TypeScript, MongoDB, AWS) Nov 2024 – Present
Digiaccel Learning Gurugram, IN

- Built a **Grafana/Prometheus** observability dashboard with **AWS Lambda**-driven staging shutdown, reducing infrastructure costs by **12.5%**.
- Reduced **API response times by 50–60%** via **Redis caching**, concurrent async processing, query optimization, and request batching.
- Established scaling baselines through **k6 load tests** on **AWS ECS Fargate**, informing peak-traffic auto-scaling and capacity policies.
- Integrated **Sentry** for error tracking and **PostHog** for product analytics, cutting **MTTR to ~1 hour** and surfacing engagement insights.
- Implemented secure direct-to-**AWS S3** uploads via **pre-signed URLs**, removing server bottlenecks and improving upload latency by **~30%**.
- Applied **BullMQ** rate-limiting and retry queues on **LeadSquared CRM** lead-sync, eliminating HTTP 429 throttling with zero lead loss.

Junior Software Engineer (MERN Stack) Dec 2023 – Nov 2024
Vilihi Virtual Services Noida, IN

- Published a shared **TypeScript Enums NPM package** unifying frontend and backend contracts, cutting type-related defects by **~20%**.
- Configured **Storybook** for isolated React component development, accelerating UI iteration, design QA, and review cycles.
- Designed **Node.js + Express** REST APIs with **MongoDB** schemas, **JWT** auth, and **Zod** validation powering core backend workflows.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript (ES6+), Python, C++, SQL

Frontend: React.js, Next.js, Redux, Tailwind CSS, MUI, HTML5, Progressive Web Apps (PWA), Storybook

Backend: Node.js, NestJS, Express.js, FastAPI, REST APIs, Microservices, WebSockets

Databases & Caching: MongoDB, MySQL, PostgreSQL, Redis, BullMQ, Celery

AI & Integrations: Claude API, OpenAI APIs, GPTZero, LLM Pipelines, Meta Graph API, Webhooks, Prompt Engineering

Cloud & DevOps: AWS (S3, ECS, Fargate, Lambda, RDS, SQS, SES, CloudWatch), Docker, Nginx, CI/CD, Vercel, Sentry, PostHog, k6

Tools & Practices: Git/GitHub, JIRA, Postman, Agile/Scrum, System Design, Code Review

KEY PROJECTS

DigiCampus | Learning Management System — Next.js, NestJS, TypeScript, MongoDB, AWS, Redis

- Delivered a production LMS powering **138+ daily active students** and **8k–12k monthly user hours** of continuous learning activity.
- Cut **API latency by 50–60%** via **Redis caching**, concurrent async processing, and strategic **MongoDB indexing** on the hottest read paths.
- Integrated **GPTZero** AI APIs to detect AI-generated content in student assessment submissions, mitigating cheating at scale.
- Converted the platform into a **Progressive Web App** with offline access, install support, and push-notification engagement.
- Optimized React with **memoization**, **code-splitting**, **lazy loading**, **debouncing**, and AbortController-based request cancellation patterns.

Narriva | AI Social Media Automation — FastAPI, PostgreSQL, Redis, Celery, AWS, Claude API

- Architected an **AI-driven social media automation** platform for content generation, publishing, customer interaction, and lead capture.
- Built **event-driven async pipelines** on **Celery + Redis** for trend detection, scheduling, lead capture, and WhatsApp alerts.
- Implemented **Meta Graph API** for Instagram/Facebook **OAuth**, media publishing, real-time **webhook** processing, and analytics.
- Engineered **FastAPI** services with **PostgreSQL**, **AWS**, **JWT** auth, and rate-limited, multi-tenant request handling at scale.
- Designed scalable **SaaS architecture** for SMB automation with secure APIs, background jobs, and third-party platform integrations.

EDUCATION

National Institute of Technology Kurukshetra 2020 – 2024
Bachelor of Technology CGPA: 7.88/10

Yaduvanshi Shiksha Niketan, Mahendragarh 2018 – 2020
Senior Secondary (Class XII) 93.2%

Yaduvanshi Shiksha Niketan, Mahendragarh 2016 – 2018
Secondary (Class X) 88.5%